BUSINESS INFORMATION DOCUMENT (BID)

AI-Powered Recruitment & Interview Intelligence Platform

Final Evaluation Assignment — AI Trainee Developer Programme

Document Attribute	Detail
Document Type	Business Information Document (BID) / Assignment Brief
Programme	AI Trainee Developer — Final Evaluation
Project Name	To be proposed by each trainee (see Section 2.3)
Mode of Work	Individual (solo) — no team collaboration permitted
Total Duration	17 working days (1st October to 27th October 2026)
Phase 1	Requirement Understanding, Clarification & WBS — 2 days
Phase 2	Core Development, Submission & Presentation — 15 days
Primary Audience	AI Trainee Developers
Secondary Audience	Trainers, Technical Mentors, Evaluation Panel
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1. Document Purpose & Audience

1.1 Purpose

This Business Information Document (BID) defines the business context, functional expectations, technical constraints, delivery timeline, and evaluation framework for the final assessment project of the AI Trainee Developer Programme.

The document serves three purposes simultaneously:

- **Business brief —** It communicates business intent rather than technical specifications. It describes what the business needs and why, leaving the "how" to the trainee.
- **Assignment contract —** It sets out mandatory deliverables, phase gates, submission standards and the weighted criteria against which work will be graded.
- **Instructional guide —** It provides guidance notes throughout to help trainees approach requirement analysis, work breakdown, prioritisation and delivery in a professional manner.

1.2 Intended Audience

Audience	How This Document Should Be Used
AI Trainee Developers	Primary audience. Read the full document before starting. Treat it as the authoritative statement of scope and acceptance. Raise ambiguities during Phase 1 rather than assuming.
Trainers & Technical Mentors	Reference for clarification sessions, checkpoint reviews, and progress tracking against the stated phase plan.
Evaluation Panel	Basis for scoring. Section 11 defines the weightage model applied at final submission and presentation.
Programme Coordinators	Reference for schedule, milestone dates, resource provisioning (sample CVs) and submission logistics.

1.3 How to Read This Document

Sections 2 to 4 establish business context and boundaries. Sections 5 and 6 define what must be built. Sections 7 and 8 define quality and technology constraints. Sections 9 to 12 define how the work will be planned, delivered, and judged.

2. Project Overview & Background

2.1 Business Context

Recruitment teams across the industry face a persistent bottleneck: the volume of applications received for any given role far exceeds the human capacity available to assess them consistently. Screening is frequently performed under time pressure, evaluation criteria drift between reviewers, and interview outcomes are recorded as unstructured notes that are difficult to compare, audit or defend.

The consequences are well understood by hiring organisations:

- Qualified candidates are missed because screening is shallow and keyword-driven.
- Different interviewers apply different standards to the same role, producing non-comparable outcomes.
- Interview content is lost; only a subjective summary survives.
- Hiring decisions cannot be traced back to evidence, creating fairness and compliance exposure.
- The cycle time from application to decision is long, and strong candidates withdraw.

2.2 The Proposed Solution

The business requires an AI-powered Recruitment & Interview Intelligence Platform that supports the hiring workflow end to end — from the moment a CV is received, through the live interview, to a consolidated, evidence-backed candidate assessment.

The platform is expected to operate across three logical stages:

Stage	Business Intent
Pre-Interview	Ingest and understand CVs, measure them objectively against defined job requirements, and produce a ranked, scored shortlist so that interviewer time is spent on the right candidates.
During-Interview	Conduct the interview through an integrated video interface while capturing the conversation as structured, speaker-attributed data instead of ad-hoc notes.
Post-Interview	Analyse what was actually asked and answered, and merge that evidence with pre-interview screening into one consolidated assessment that a hiring manager can act on.

The platform is intended to augment human decision-making, not to replace it. Every AI-generated score, ranking or flag must be explainable and traceable to the underlying evidence.

2.3 Project Naming — Trainee Responsibility

No project name is assigned in this document. This omission is deliberate.

Each trainee must propose their own project name and product identity as a graded deliverable. The name is not a cosmetic exercise; it is treated as a signal of product thinking. Trainers will assess whether the trainee understood the problem well enough to articulate what the product is and who it serves.

The naming deliverable must include:

- A proposed product name.
- A one-line positioning statement (what it does, for whom).
- A short rationale of two to four sentences explaining the reasoning behind the name.
- Consistent application of the name across the repository, README, application interface and final presentation.

3. Objectives & Goals

3.1 Business Objectives

#	Objective	Success Indicator
B1	Reduce manual CV screening effort	A batch of CVs can be screened and ranked against a job requirement without manual review of each document.
B2	Improve consistency of candidate evaluation	The same candidate and job requirement produce a stable, reproducible score with a visible breakdown.
B3	Capture interview content as structured data	Every interview produces a speaker-attributed transcript rather than free-form notes.
B4	Enable evidence-based hiring decisions	Each assessment cites the CV content and transcript segments that justify its conclusions.
B5	Shorten time-to-decision	A consolidated assessment is available immediately after the interview concludes.

3.2 Learning & Evaluation Objectives

Beyond the product itself, this assignment is designed to assess the following competencies:

#	Competency	How It Is Demonstrated
L1	Requirement analysis	Quality and depth of clarification questions and documented assumptions in Phase 1.
L2	Planning & estimation	Realism, granularity and traceability of the Work Breakdown Structure.
L3	Agentic AI engineering	Appropriate use of LangGraph for multi-step, stateful reasoning rather than single monolithic prompts.
L4	Backend & API design	Clean, documented FastAPI service boundaries and data contracts.
L5	System integration	Ability to combine video, audio, transcription, LLM reasoning and persistence into one coherent application.
L6	Delivery discipline	Dockerised, reproducible build; clean repository; meaningful commit history.
L7	Creativity & initiative	Originality and value of the self-proposed innovation feature and product name.
L8	Communication	Clarity of documentation and the final presentation and demonstration.

3.3 Definition of Success

A submission is considered successful when all of the following hold:

- All six Core (Mandatory) features are implemented and demonstrable end to end.
- One original Creative / Innovation feature is implemented with a written rationale.
- The application runs from a clean machine using documented Docker commands, with no undocumented manual setup.
- The GitHub repository contains a complete README, clear setup instructions and an incremental commit history.
- A proposed project name and product positioning are presented and applied consistently.
- The trainee can explain and defend their architectural decisions during the presentation.

4. Scope

4.1 In-Scope

Area	Included in Scope
Job Requirement Definition	Ability to define or configure a job requirement profile — required skills, qualifications, minimum experience and selection criteria.
Job Post	We are looking for a talented and motivated professional to join our team. If you are interested in this opportunity, please send your CV to the contact email provided in the job post. Feel free to share this opportunity with anyone who may be interested.
CV Ingestion & Parsing	Upload and parse candidate CVs (minimum: PDF; DOCX support encouraged) and extract structured attributes.
Screening & Ranking	Automated matching of parsed CVs against the job requirement profile, with an ordered candidate ranking.
Candidate Scoring	Objective, criteria-weighted suitability scoring with a visible breakdown per criterion.
Video Interview	An integrated video-chat interface supporting a real-time interviewer–candidate session with concurrent audio capture.
Speech Recognition	Transcription of both participants with speaker attribution and a structured, timestamped transcript.
Q&A Analysis	Analysis of interviewer questions and candidate responses for relevance, accuracy, completeness and quality.
Consolidated Assessment	A unified candidate report combining screening, scoring and interview analysis.
Innovation Feature	One original, trainee-conceived feature with documented rationale.
Packaging & Delivery	Dockerised application, GitHub repository, README, setup documentation and final presentation.

4.2 Out-of-Scope

The following are explicitly excluded. Trainees must not spend effort here; doing so will be treated as a scope-management failure.

Area	Excluded — Rationale
Production deployment	No cloud hosting, CI/CD pipelines, load balancing or infrastructure-as-code required. Local Docker execution is sufficient.

Area	Excluded — Rationale
Enterprise authentication	No SSO, LDAP, OAuth provider integration or enterprise identity management. Simple role separation is acceptable.
ATS / HRIS integration	No integration with commercial applicant tracking or HR information systems.
Payroll, offer & onboarding	The workflow ends at the consolidated candidate assessment.
Job posting & sourcing	No job-board publishing, candidate sourcing or application-portal functionality.
Interview scheduling & calendaring	Calendar integration and automated scheduling are excluded unless a trainee elects it as their innovation feature.
Multi-tenancy & billing	No tenant isolation, subscription management or commercial licensing logic.
Native mobile applications	Web-based delivery only.
Custom model training	Training or fine-tuning foundation models from scratch is out of scope. Use of existing models and APIs is expected.
Formal accessibility certification	WCAG conformance auditing is not required, though sensible accessibility practice is welcome.

4.3 Scope Boundary Guidance

> **Guidance Note — Protecting Your Scope**
>
> Seventeen development days is less time than it appears once integration and debugging are accounted for. The most common failure mode is over-building the first feature and running out of time before the interview pipeline is working.
>
> Build a thin end-to-end slice first — CV in, interview conducted, assessment out — then deepen each stage. A shallow but complete pipeline is worth substantially more marks than one beautifully engineered module and four unfinished ones.

5. Functional Features

Features are grouped by the stage of the recruitment workflow in which they operate. Within each stage, Core (Mandatory) features must be delivered; Stretch (Bonus) features are optional and carry additional credit.

5.1 Feature Summary

#	Feature	Stage	Priority	Classification
F1	CV Screening & Requirement-Based Candidate Ranking	Pre-Interview	Must Have	Core
F2	AI-Based Candidate Scoring	Pre-Interview	Must Have	Core
F3	AI-Powered Video Interview	During-Interview	Must Have	Core
F4	Real-Time Speech Recognition & Transcription	During-Interview	Must Have	Core
F5	AI-Based Interview Question & Answer Analysis	Post-Interview	Must Have	Core
F6	Consolidated Candidate Assessment	Post-Interview	Must Have	Core
F7	Candidate Identification & Real-Time Profile Overview	During-Interview	Could Have	Stretch
F8	CV Claim Verification Through Interview Analysis	Post-Interview	Could Have	Stretch
F9	Comprehensive Candidate Interview Scoring	Post-Interview	Could Have	Stretch
F10	Interview Quality Assessment	Post-Interview	Could Have	Stretch
F11	Trainee-Proposed Innovation Feature	Any	Must Have	Creative

5.2 Stage 1 — Pre-Interview

This stage covers everything that happens before a candidate enters an interview: ingesting CVs, understanding the job requirement, and producing a ranked and scored shortlist.

5.2.1 Core (Mandatory) Features

F1 — CV Screening & Requirement-Based Candidate Ranking

The system must automatically analyse and screen submitted CVs against a defined set of job requirements and rank candidates by overall suitability.

Expected behaviour:

- Accept a job requirement profile covering required and preferred skills, qualifications, minimum years of experience and any additional selection criteria.
- Ingest and parse multiple CVs in a single batch and extract structured information — contact details, education, work history, skills, certifications, project experience.
- Match extracted attributes against the requirement profile using semantic understanding, not literal keyword matching alone. A CV stating "built REST services in Django" should register against a "Python web development" requirement.
- Produce an ordered ranking of all screened candidates with a clear indication of relative suitability.
- Expose the reasoning: for each candidate, indicate which requirements were met, partially met or unmet, with supporting evidence drawn from the CV.
- Handle imperfect input gracefully — non-standard layouts, missing sections, multi-column formats and scanned or image-heavy documents should fail safely rather than crash the batch.

F2 — AI-Based Candidate Scoring

The system must evaluate each candidate against predefined requirements and generate an objective suitability score.

Expected behaviour:

- Score across distinct dimensions at minimum: technical skills, relevant experience, and academic or professional qualifications. Additional dimensions may be defined by the trainee.
- Support configurable weighting per dimension so that the profile can be tuned per role.
- Present both an aggregate score and the per-dimension breakdown that produced it.
- Accompany each dimension score with a short natural-language justification.
- Produce stable, reproducible results — the same CV and requirement profile should not yield materially different scores on repeated runs.
- Persist scores so that they can be retrieved later and merged into the consolidated assessment.

5.2.2 Stretch (Bonus) Features — Pre-Interview

No stretch features are defined for this stage. Trainees who wish to extend the pre-interview stage should do so through their Innovation Feature (Section 6).

5.3 Stage 2 — During-Interview

This stage covers the live interview session: the video interaction itself and the real-time capture and structuring of the conversation.

5.3.1 Core (Mandatory) Features

F3 — AI-Powered Video Interview

The system must allow interviews to be conducted through an integrated video-chat interface that enables real-time communication while continuously capturing audio and conversation data for analysis.

Expected behaviour:

- Provide a browser-based video and audio session between interviewer and candidate.
- Support the basic session lifecycle: create or schedule a session, join, conduct, and end.

- Capture the audio stream continuously throughout the session and route it into the analysis pipeline.
- Associate every session with a specific candidate and a specific job requirement profile.
- Provide standard in-call controls — mute, camera toggle, end session.
- Degrade gracefully on network interruption; a dropped frame must not terminate the recording or lose the transcript captured so far.
- Persist the session record, including start and end times and participants.

F4 — Real-Time Speech Recognition & Transcription

The system must capture and transcribe the speech of both interviewer and interviewee, distinguish between speakers, and generate a structured interview transcript.

Expected behaviour:

- Transcribe speech during the session with minimal perceptible lag.
- Attribute each utterance to the correct speaker (speaker diarisation or channel separation).
- Timestamp each utterance so that transcript segments can be referenced later as evidence.
- Produce a structured transcript — a sequence of speaker-attributed, timestamped turns — rather than an undifferentiated block of text.
- Persist the transcript and make it retrievable after the session ends.
- Handle accented speech, overlapping speech and silence without failing; imperfect transcription is acceptable provided the system remains stable.

5.3.2 Stretch (Bonus) Features — During-Interview

F7 — Candidate Identification & Real-Time Profile Overview

Identify the interviewee via facial and/or voice recognition, retrieve their stored information, and present a concise profile overview within the interview interface.

- Match the person joining the session against the candidate records already in the system.
- On a confident match, surface a compact profile panel to the interviewer — name, headline experience, key skills, pre-interview score and flagged areas to probe.
- Handle non-identification and low-confidence matches explicitly; never silently assume an identity.
- Treat biometric data with particular care — see Section 7.3 on candidate data privacy. Consent handling must be visible in the flow.

5.4 Stage 3 — Post-Interview

This stage converts captured interview data into analysis, and merges that analysis with pre-interview screening into a single decision-support artefact.

5.4.1 Core (Mandatory) Features

F5 — AI-Based Interview Question & Answer Analysis

The system must analyse interviewer questions and candidate responses to determine how effectively the candidate addressed the job requirements.

Expected behaviour:

- Segment the transcript into question–answer pairs, correctly associating each response with the question that prompted it.
- Evaluate each response on, at minimum: relevance to the question asked, factual and technical accuracy, completeness, and overall quality of articulation.
- Map questions and answers back to the specific job requirements they address, so coverage gaps become visible.
- Provide a short written justification for each evaluation, referencing the transcript segment that supports it.
- Handle follow-up questions, clarifications and multi-part answers sensibly.
- Identify requirements that were never explored during the interview.

F6 — Consolidated Candidate Assessment

The system must combine CV screening, requirement-based scoring and interview analysis into a single consolidated, structured assessment supporting consistent, data-driven hiring decisions.

Expected behaviour:

- Merge pre-interview screening results, candidate scores and post-interview Q&A analysis into one coherent candidate report.
- Present an overall suitability position supported by component scores, not a single opaque number.
- Summarise strengths, weaknesses, evidence gaps and recommended areas for further verification.
- Make the report exportable or printable in a readable format (PDF, structured HTML or equivalent).
- Support side-by-side comparison of multiple assessed candidates for the same role.
- Maintain traceability throughout — every conclusion should be attributable to CV content or a transcript segment.

5.4.2 Stretch (Bonus) Features — Post-Interview

F8 — CV Claim Verification Through Interview Analysis

- Cross-reference claims made in the CV against what the candidate said during the interview.
- Flag inconsistencies, contradictions and claims that were asserted but never substantiated.
- Distinguish clearly between a genuine contradiction, an unverified claim and a claim that was simply never discussed.
- Present flags as prompts for human follow-up, never as accusations or automated rejections.

F9 — Comprehensive Candidate Interview Scoring

- Generate an overall interview performance score distinct from the CV-based score.
- Score across technical competency, communication, relevance of responses, problem-solving approach, depth of experience demonstrated, and alignment with job requirements.
- Provide the per-dimension breakdown and supporting evidence.

F10 — Interview Quality Assessment

- Evaluate the quality of the interview process itself, not only the candidate.

- Assess relevance and effectiveness of the interviewer questions, coverage of the required competencies, consistency of process across candidates, and overall interaction quality.
- Recommend improvements — competencies left unexplored, questions that produced little signal, imbalanced speaking time.

6. Creative / Innovation Feature (Trainee-Independent Task)

6.1 Purpose of This Task

Every trainee must propose and implement ONE original feature of their own that meaningfully enhances the platform.

This task is deliberately unspecified. The features in Section 5 tell you what to build; this section asks you to decide what should be built. It is included because engineering judgement — identifying a real gap, evaluating whether it is worth solving, and delivering it within constraints — cannot be assessed through prescribed requirements.

Because every trainee will propose something different, this task also gives trainers a clear, uncontaminated view of individual thinking and initiative.

6.2 Requirements

#	Requirement	Detail
C1	Originality	The feature must be the trainee's own idea. It must not be one of features F1–F10, nor a cosmetic variation, subset or renaming of them.
C2	Relevance	It must enhance the recruitment and interview intelligence platform. An unrelated capability bolted on will not be credited.
C3	Written rationale	A short document (approximately 300–500 words) stating the problem it solves, who benefits, the value it adds, and why it was prioritised over alternatives the trainee considered.
C4	Independent design	Designed and implemented by the trainee alone, without trainer-provided specification.
C5	Working implementation	It must be implemented and demonstrable in the running application. A concept or mock-up alone does not satisfy this requirement.
C6	Integration	It must integrate coherently with the rest of the platform rather than existing as a disconnected side module.
C7	Documentation	It must be documented in the README, clearly identified as the Innovation Feature.
C8	Presentation	It must be demonstrated and justified during the final presentation.

6.3 Assessment Dimensions

Dimension	What Trainers Will Look For
Problem identification	Does the feature address a real gap in the recruitment workflow, or a hypothetical one?
Originality	Is the thinking genuinely the trainee's own, or an obvious extension of the given list?
Value articulation	Can the trainee explain who benefits and why, in business terms rather than technical terms?
Prioritisation judgement	Did the trainee weigh alternatives and choose deliberately, or pick the first idea that occurred?
Technical execution	Is it actually working, reasonably built and integrated?
Scope discipline	Was it delivered without compromising the Core features?

6.4 Constraints

- Exactly one innovation feature is required. Additional features will not attract additional marks and may indicate poor prioritisation.
- It must not be attempted at the expense of Core features.
- It must respect all non-functional requirements in Section 7, particularly candidate data privacy.
- It must operate within the technical constraints in Section 8.

7. Non-Functional Requirements

7.1 Performance & Responsiveness

#	Requirement	Expectation
N1	CV parsing throughput	A batch of 20 CVs should be parsed, screened, scored and ranked within a reasonable, documented time window. State your measured figure in the README.
N2	Transcription latency	Transcribed text should appear with minimal perceptible delay during the live session.
N3	Video session quality	The video and audio session must remain usable and stable for the duration of a typical interview.
N4	API responsiveness	Non-AI API endpoints should respond quickly. Long-running AI operations must be asynchronous, with progress or status feedback rather than a blocked request.
N5	Concurrency	The system should tolerate at least one active interview session alongside background CV processing without failure.
N6	Graceful degradation	Failure of an external AI service must produce a clear error state, not a crash or silent data loss.

7.2 Security

#	Requirement	Expectation
N7	Credential management	All API keys, secrets and connection strings must be supplied via environment variables. No secret may ever be committed to the repository. Provide a .env.example with placeholder values.
N8	Input validation	All API inputs must be validated. Uploaded files must be checked for type and size before processing.
N9	Access separation	Interviewer and candidate capabilities must be separated. A candidate must not be able to access scoring, ranking or assessment data.
N10	Transport security	Where media streams and APIs are exposed, use secure transport. Document any local-development exceptions.
N11	Dependency hygiene	Pin dependency versions. Avoid unmaintained or unnecessary packages.
N12	Error disclosure	Error responses must not leak stack traces, file paths, internal configuration or prompt content to the client.

7.3 Candidate Data Privacy

This section is mandatory and will be assessed. CVs and interview recordings contain personal data — names, contact details, addresses, employment history, education records, voice and facial imagery. Trainees are expected to handle this data responsibly, exactly as they would in a production hiring system.

#	Requirement	Expectation
N13	Use mock or anonymised data	Use only the 20 provided sample CVs, or synthetic data you create. Never use real CVs belonging to real individuals who have not consented.
N14	No personal data in the repository	Do not commit CVs, recordings, transcripts, screenshots containing personal details, or database dumps. Add these paths to .gitignore.
N15	Anonymise demonstration material	Redact or replace identifying details in screenshots, README examples, presentation slides and logs.
N16	Data minimisation	Store only what the feature genuinely requires. Do not retain raw audio or video beyond what is needed to produce the transcript, unless justified.
N17	Biometric caution	If implementing F7 (facial or voice identification), treat biometric data as the most sensitive category. Make consent explicit in the flow and state clearly in the README how the data is stored and for how long.
N18	Third-party disclosure	Document in the README which external services receive candidate data (LLM providers, transcription APIs) and what is transmitted to each.
N19	Deletion capability	Provide a means of deleting a candidate and their associated data. A documented administrative endpoint or script is sufficient.
N20	Fairness awareness	Scoring must be based on job-relevant criteria only. Do not use, infer or store protected characteristics such as age, gender, ethnicity, nationality, religion or marital status as scoring inputs. Note any fairness limitations of your approach in the README.

7.4 Reliability, Maintainability & Usability

#	Requirement	Expectation
N21	Reproducibility	The application must start reliably on a clean machine using only the documented Docker commands.
N22	Observability	Implement structured logging for key operations and failures. Logs must not contain personal data or secrets.
N23	Code quality	Consistent structure, meaningful naming, separation of concerns, and no large blocks of commented-out or dead code.
N24	Documentation	README covering purpose, architecture, setup, configuration, usage and known limitations. FastAPI endpoints documented via

#	Requirement	Expectation
		OpenAPI.
N25	Usability	The interface must be navigable without verbal explanation. Long-running operations must show progress. Errors must be understandable to a non-technical user.
N26	Data persistence	Candidate records, scores, transcripts and assessments must survive an application restart.

8. Technical Requirements & Constraints

8.1 Mandatory Technology Stack

The following are non-negotiable. Deviation requires written approval from a trainer during Phase 1 and must be recorded in the clarification document.

Layer	Mandated Technology	Requirement
Backend language	Python	All backend and AI logic must be written in Python. A recent stable version (3.10 or later) is recommended.
Agent / orchestration	LangGraph	LangGraph is the preferred and expected framework for agent implementation and orchestration. Multi-step reasoning workflows should be modelled as explicit graphs with defined state, nodes and transitions — not as a chain of ad-hoc prompt calls.
API framework	FastAPI	All backend services must be exposed through FastAPI, with Pydantic models for request and response validation and automatic OpenAPI documentation.
Containerisation	Docker	The complete application must be containerised. Multi-service setups must be orchestrated with Docker Compose.
Source control	GitHub	The project must be submitted as a GitHub repository with a complete, incremental commit history.
Frontend	Trainee's choice	Any framework or approach may be used. The choice must be justified in the README and in the presentation.

8.2 LangGraph Usage Expectations

LangGraph is mandated because the problem is genuinely multi-stage and stateful. Trainers will specifically assess whether it has been used meaningfully.

- Model reasoning workflows as graphs with clearly defined state, nodes and edges.
- Use it where it adds value — CV analysis pipelines, interview response evaluation, consolidated assessment generation.
- Maintain explicit state across steps rather than rebuilding context from scratch on each call.
- Implement error handling and retry paths within the graph.
- Be prepared to explain your graph topology, why each node exists, and what state flows between them.

Wrapping a single LLM call in a one-node graph to satisfy the requirement on paper will be identified and will not be credited.

8.3 Dockerisation Requirements

#	Requirement	Detail
D1	Complete containerisation	Backend, frontend and any supporting services (database, cache, vector store) must run in containers.
D2	Single-command startup	The reviewer must be able to bring the full application up with one documented command.
D3	Compose orchestration	Use Docker Compose to define services, networks, volumes and dependencies.
D4	Environment configuration	All configuration via environment variables. Ship a .env.example; never ship a populated .env.
D5	Persistent volumes	Use named volumes so that data survives container restarts.
D6	Clean-machine verification	Before submission, verify startup on a machine with no prior project artefacts or cached local dependencies.
D7	Image discipline	Use appropriate base images and a .dockerignore. Avoid unnecessarily large images.
D8	Documented commands	The README must list every command required to build, run, stop and reset the application.

8.4 GitHub Repository Requirements

#	Requirement	Detail
G1	Repository access	A GitHub repository shared with the trainer team before the submission deadline.
G2	README completeness	Project name, positioning statement, feature list (Core, Stretch attempted, Innovation), architecture overview, technology stack with justification, prerequisites, setup and run instructions, configuration reference, usage walkthrough, privacy notes, and known limitations.
G3	Commit history	Incremental, meaningful commits made throughout the 17 development days. A single bulk commit at the end is a significant negative signal.
G4	Commit messages	Descriptive and specific. "Implement speaker diarisation for transcript segmentation" — not "update", "fix" or "wip".
G5	Branching	A sensible branching approach is encouraged. Keep the default branch in a working state.
G6	Repository hygiene	A proper .gitignore. No secrets, no candidate data, no virtual environments, no build artefacts, no IDE configuration.
G7	Project documentation	The clarification document, WBS and innovation-feature rationale must be committed to the repository, ideally under a /docs

#	Requirement	Detail
		directory.
G8	Final tag	Tag the final submission commit (for example, v1.0-final) so the evaluated state is unambiguous.

8.5 Sample Data Provision

Twenty sample CVs will be provided for development, testing and demonstration purposes.

- The sample set is representative of realistic input, including varied formats, layouts and levels of completeness.
- All demonstrations during the final presentation must use this sample set unless otherwise agreed.
- Trainees may create additional synthetic CVs to test edge cases; these should be committed under a clearly marked test-data directory.
- The provided CVs must not be committed to a public repository. Follow requirement N14.

8.6 Recommended but Optional

The following are suggestions, not requirements. Trainees are free to choose alternatives.

- Relational or document database for candidate, session and assessment persistence.
- Vector store for semantic CV and requirement matching.
- WebRTC-based approach for the video interview interface.
- A managed or self-hosted speech-to-text service for transcription and diarisation.
- Background task handling for long-running AI operations.
- Automated tests for core business logic.

9. Project Phases & Timeline

The assignment runs for 20 working days across two phases. The day-wise plan below is indicative and represents the trainer's expected pacing. Trainees are expected to produce their own detailed plan in Phase 1 and may adjust sequencing — but the phase boundaries and the final submission date are fixed.

9.1 Phase Summary

Phase	Title	Duration	Primary Outcome
Phase 1	Requirement Understanding, Clarification & WBS Preparation	Days 1–3	Approved clarification document, proposed project name and a detailed Work Breakdown Structure.
Phase 2	Core Development, Submission & Presentation	Days 4–20	Working Dockerised application, GitHub repository, final submission and presentation.

9.2 Phase 1 — Detailed Plan (Days 1–3)

Day	Focus	Key Activities	Expected Output
Day 1	Requirement study & domain research	Read this BID end to end. Research the recruitment and interview domain. Identify ambiguous, incomplete or conflicting requirements. Survey candidate technologies for video, transcription and CV parsing. Draft the clarification question list.	Draft clarification questions; initial technology research notes.
Day 2	Clarification & solution conceptualisation	Raise clarification questions with the trainer. Record answers and any agreed assumptions. Draft the high-level architecture and LangGraph agent design. Finalise the technology stack. Propose the project name and positioning.	Clarification document with trainer responses; high-level architecture; proposed project name and rationale.
Day 3	Work breakdown & planning	Decompose the scope into work packages and tasks. Estimate effort per task. Sequence the 17 development days. Identify risks, dependencies and mitigations. Initialise the GitHub repository with structure and README skeleton.	Work Breakdown Structure; 17-day delivery plan; risk register; initialised repository. Phase 1 gate review.

9.3 Phase 2 — Detailed Plan (Days 4–20)

Days	Focus	Key Activities	Expected Output
Days 4–5	Foundation & environment setup	Repository structure. FastAPI application skeleton with health endpoint. Dockerfile and Docker Compose. Database schema and models. CV upload and parsing pipeline. Base LangGraph graph and state definition.	Running containerised skeleton; CV ingestion working.
Days 6–8	Pre-Interview core (F1, F2)	Job requirement profile definition. Semantic CV-to-requirement matching. Candidate ranking logic. Multi-dimensional scoring with configurable weights. Score justification generation. Screening and ranking UI.	F1 and F2 complete and demonstrable on the 20 sample CVs.
Days 9–11	During-Interview core, part 1 (F3)	Video-chat interface. Session lifecycle management. Real-time audio and video streaming. Continuous audio capture. Session persistence and linkage to candidate and job profile.	F3 complete; a live interview session can be conducted end to end.
Days 12–13	During-Interview core, part 2 (F4)	Speech-to-text integration. Speaker diarisation or channel separation. Timestamped, structured transcript generation. Live transcript display. Transcript persistence and retrieval.	F4 complete; speaker-attributed transcripts produced and stored.
Days 14–15	Post-Interview core (F5, F6)	Question–answer pair segmentation. Response evaluation across relevance, accuracy, completeness and quality. Requirement coverage mapping. Consolidated assessment generation. Report view and export. Candidate comparison.	F5 and F6 complete; all six Core features working end to end.
Days 16–17	Creative / Innovation feature (F11)	Finalise the innovation concept. Write the rationale document. Design, implement and integrate the feature. Test it against the sample data set.	F11 implemented and documented; rationale committed to /docs.
Day 18	Stretch features & hardening	Attempt one or more stretch features only if all Core features are complete. Otherwise, harden Core: error handling, logging, privacy controls, input validation, UI polish.	Optional F7–F10; hardened, stable application.
Day 19	Integration, testing & documentation	Full end-to-end testing with all 20 sample CVs. Clean-machine Docker	Release candidate; complete

Days	Focus	Key Activities	Expected Output
		verification. README completion. Architecture diagram. Anonymise all demonstration material. Prepare presentation and demo script.	documentation; rehearsed demo.
Day 20	Final submission & presentation	Final bug fixes and freeze. Tag the release. Submit the repository. Deliver the presentation: problem framing, project name and positioning, architecture walkthrough, live demonstration, innovation feature justification, lessons learned, Q&A.	Final submission; presentation delivered.

9.4 Milestones & Gates

Milestone	Description	Gate Criteria
M1 — End of Day 3	Phase 1 gate review	Clarification document, project name proposal and WBS submitted and reviewed. Repository initialised.
M2 — End of Day 8	Pre-Interview checkpoint	F1 and F2 demonstrable against the sample CV set.
M3 — End of Day 13	Interview pipeline checkpoint	F3 and F4 demonstrable; a full interview can be conducted and transcribed.
M4 — End of Day 15	Core feature completion	All six Core features working end to end.
M5 — End of Day 17	Innovation feature complete	F11 implemented, integrated and documented with rationale.
M6 — End of Day 19	Release candidate	Clean-machine Docker verification passed; documentation complete.
M7 — Day 20	Final submission & presentation	Repository tagged and submitted; presentation delivered.

10. Deliverables

All deliverables listed below are mandatory unless explicitly marked optional.

10.1 Phase 1 Deliverables (Due End of Day 3)

#	Deliverable	Description & Acceptance
P1.1	Proposed project name & positioning	Product name, one-line positioning statement and a two-to-four-sentence rationale. Must be applied consistently thereafter.
P1.2	Requirement clarification document	A structured list of ambiguities identified, questions raised, trainer responses received, and assumptions adopted where no answer was available. Quality of the questions is assessed, not merely their quantity.
P1.3	Work Breakdown Structure (WBS)	Hierarchical decomposition to task level with effort estimates, dependencies, sequencing across the 17 development days, and a definition of done per task.
P1.4	High-level architecture outline	System component diagram, proposed technology stack with justification, and the intended LangGraph agent design.
P1.5	Risk register	Identified technical and schedule risks with likelihood, impact and mitigation approach.
P1.6	Initialised GitHub repository	Repository created, structured, shared with trainers, with a README skeleton and a working .gitignore.

10.2 Phase 2 Deliverables (Due End of Day 20)

#	Deliverable	Description & Acceptance
P2.1	Working Dockerised application	Complete application running via documented Docker commands, verified on a clean machine, with all six Core features and the Innovation feature functional.
P2.2	GitHub repository	Final repository with complete source code, incremental commit history, and a tagged final release commit.
P2.3	README documentation	Complete as specified in requirement G2.
P2.4	Innovation feature rationale	The 300–500-word rationale document described in requirement C3, committed under /docs.
P2.5	Updated WBS & plan	The evolved WBS showing actual versus planned progress, committed under /docs.
P2.6	Architecture documentation	Final architecture diagram and a description of the LangGraph agent design as built.

#	Deliverable	Description & Acceptance
P2.7	Privacy & data handling note	A short statement covering which external services receive candidate data, what is stored, retention approach and known fairness limitations.
P2.8	Final presentation	A 20–30 minute session: problem framing, project name and positioning, architecture walkthrough, live demonstration using the 20 sample CVs, innovation feature justification, challenges and lessons learned, followed by Q&A.
P2.9	Presentation deck	Supporting slides, with all demonstration material anonymised.

10.3 Submission Standards

- All deliverables must be submitted by the end of Day 20. Late submissions attract a penalty at the panel's discretion.
- Documentation deliverables must be committed to the repository under /docs, not sent separately.
- The application must run without trainee intervention. "It works on my machine" is not an acceptable outcome.
- No candidate personal data may be present in the submitted repository.
- The trainee must be able to explain any part of the submitted codebase on request.

11. Evaluation Criteria & Weightage

The submission is scored out of 100 marks, with additional bonus credit available for stretch features. The weighting reflects the programme's priorities: disciplined delivery of the mandatory scope, sound engineering, and demonstrated independent thinking.

11.1 Overall Weightage

#	Evaluation Area	Weightage	What Is Assessed
E1	Phase 1 — Requirement Understanding, Clarification & WBS	10%	Depth of requirement analysis, quality of clarification questions, realism and granularity of the WBS, risk identification and architecture thinking.
E2	Core Features — Pre-Interview (F1, F2)	15%	CV screening accuracy, semantic matching quality, ranking logic, scoring dimensions, weighting configurability and justification transparency.
E3	Core Features — During-Interview (F3, F4)	15%	Video session stability, audio capture reliability, transcription quality, speaker attribution accuracy and transcript structure.
E4	Core Features — Post-Interview (F5, F6)	15%	Q&A segmentation, evaluation quality, requirement coverage mapping, consolidation logic and traceability of conclusions to evidence.
E5	Creative / Innovation Feature (F11)	10%	Originality, problem relevance, quality of written rationale, prioritisation judgement, implementation quality and integration.
E6	Proposed Project Name & Product Thinking	5%	Appropriateness and memorability of the name, clarity of positioning, quality of rationale and consistency of application across repository, UI, docs and presentation.
E7	Technical Implementation Quality	10%	Meaningful LangGraph usage, FastAPI design, architecture, code quality, separation of concerns and error handling.
E8	Dockerisation & Repository Quality	8%	Single-command startup, clean-machine reproducibility, Compose configuration, commit history quality, repository hygiene and README completeness.
E9	Non-Functional Requirements	5%	Performance, security, and in particular candidate data privacy handling and

#	Evaluation Area	Weightage	What Is Assessed
			fairness awareness.
E10	Final Presentation & Demonstration	7%	Clarity of communication, quality of live demo, ability to justify decisions and handle technical questions.
	TOTAL	**100%**	

11.2 Bonus Marks — Stretch Features

Stretch features are credited only when all six Core features are complete and working. Bonus marks are capped at 10 marks in total.

#	Stretch Feature	Bonus	Condition
F7	Candidate Identification & Real-Time Profile Overview	Up to +3	All Core features complete; consent and biometric privacy handled (N17).
F8	CV Claim Verification Through Interview Analysis	Up to +3	All Core features complete; flags framed as human follow-up prompts.
F9	Comprehensive Candidate Interview Scoring	Up to +2	All Core features complete; per-dimension breakdown provided.
F10	Interview Quality Assessment	Up to +2	All Core features complete; actionable recommendations produced.
	MAXIMUM BONUS	**+10**	

11.3 Feature-Level Scoring Rubric

Each feature is assessed against the following scale, applied proportionally within its weighting band.

Rating	Band	Description
Excellent	90–100%	Fully functional, robustly handled edge cases, well integrated, clearly explainable, evidence-backed output.
Good	75–89%	Fully functional for the main path with minor gaps in edge-case handling or transparency.
Satisfactory	60–74%	Functional but limited — works on happy path only, shallow analysis or weak justification.
Partial	40–59%	Implemented but unreliable, incomplete or poorly integrated with

Rating	Band	Description
		the rest of the platform.
Minimal	1–39%	Attempted but not working, or present only as a stub or mock.
Not delivered	0%	Absent from the submission.

11.4 Deductions

Condition	Impact
Application does not start from a clean machine using documented commands	Up to −10 marks
Secrets, credentials or API keys committed to the repository	Up to −8 marks
Candidate personal data committed to the repository	Up to −8 marks
Single bulk commit or absent commit history	Up to −5 marks
Stretch features attempted while Core features remain incomplete	Up to −5 marks (scope management)
Innovation feature duplicates or trivially varies a listed feature	Innovation marks (E5) forfeited
Evidence of work not performed by the trainee, or inability to explain submitted code	Referral to the panel; may result in disqualification
Late submission	At panel discretion

11.5 What Distinguishes a Strong Submission

- Complete Core delivery over partial breadth — all six mandatory features working beats a scattering of half-built ones.
- Explainable AI output — every score and flag traceable to evidence rather than emerging from an opaque prompt.
- Genuine LangGraph usage — stateful, multi-node reasoning the trainee can walk through and defend.
- Effortless setup — the reviewer runs one command and the application works.
- Visible thinking — a clarification document and WBS that show the problem was genuinely understood.
- Independent judgement — an innovation feature that addresses a real gap, with a rationale that convinces.
- Honest limitations — a README that states clearly what does not work and why, rather than overstating.

12. Assumptions & Constraints

12.1 Assumptions

#	Assumption
A1	Trainees have working knowledge of Python, REST API concepts and basic containerisation, and can independently research unfamiliar libraries.
A2	Trainees have access to a development machine capable of running Docker and a local video session.
A3	Access to the required AI and speech-to-text services (or approved alternatives) will be available for the duration of the assignment.
A4	Twenty sample CVs will be provided at the start of Phase 1 and are representative of realistic recruitment input.
A5	A trainer will be available during Phase 1 to respond to clarification questions within a reasonable turnaround.
A6	Interviews demonstrated during evaluation will be simulated, using mock candidates and the provided sample data.
A7	Transcription and analysis are expected in English unless otherwise agreed during Phase 1.
A8	Imperfect AI output is acceptable and expected. The system is assessed on design, transparency and robustness, not on achieving benchmark-grade model accuracy.
A9	A single concurrent interview session is sufficient for demonstration purposes.
A10	Trainees will raise blockers promptly rather than losing days to a problem that a short conversation could resolve.

12.2 Constraints

#	Constraint	Detail
CN1	Fixed duration	20 working days total. Phase boundaries and the Day 20 submission date are fixed and will not be extended.
CN2	Solo delivery	Work must be performed individually. Collaboration between trainees on design or code is not permitted. General discussion of concepts is acceptable; sharing implementations is not.
CN3	Mandated stack	Python, LangGraph, FastAPI and Docker are non-negotiable without documented trainer approval.
CN4	Local execution only	The application must run locally via Docker. Cloud deployment is neither required nor credited.

#	Constraint	Detail
CN5	Sample data only	Development and demonstration must use the provided sample CVs or trainee-generated synthetic data. Real personal data is prohibited.
CN6	Privacy compliance	All requirements in Section 7.3 are mandatory and assessed.
CN7	External service dependency	Reliance on third-party AI services introduces cost, rate-limit and availability risk. Trainees must design for failure and document any limits encountered.
CN8	Scope discipline	Out-of-scope work (Section 4.2) will not be credited and may be penalised as poor prioritisation.
CN9	Originality	The innovation feature must be the trainee's own. AI coding assistants may be used as tools, but the trainee must fully understand and be able to explain every part of the submission.
CN10	Demonstrability	Any feature that cannot be demonstrated live during the presentation will be treated as not delivered.

12.3 Known Risks

Risk	Likelihood	Suggested Mitigation
Real-time video and audio pipeline proves harder than estimated	High	Prototype the riskiest component early, during Days 4–5, rather than deferring it to Day 9.
Speaker diarisation accuracy is poor	Medium	Consider separate audio channels per participant instead of relying on diarisation from a mixed stream.
CV parsing fails on non-standard layouts	High	Design for graceful failure from the outset; log and skip rather than aborting the batch.
External AI service rate limits or cost caps reached	Medium	Cache aggressively during development; batch requests; avoid re-running full pipelines unnecessarily.
Over-engineering the first feature	High	Deliver a thin end-to-end slice first, then deepen. Track progress against the WBS daily.
Docker works locally but not on a clean machine	Medium	Verify on a clean environment by Day 19 at the latest, not on Day 20.

13. AI Usage Guidelines

This section defines where AI tools may and may not be used during the project. These guidelines apply to both phases and must be followed by every trainee.

13.1 Permitted Use of AI

Trainees may use AI to support the following activities:

#	Activity
AI1	Coding
AI2	UI design
AI3	Paid LLM or local LLM
AI4	Speech-to-text
AI5	Video feed collection

13.2 Activities Without AI Support

The following activities must be completed by the trainee without any support from AI:

#	Activity
NA1	Work Breakdown Structure (WBS)
NA2	Software Requirements Specification (SRS)
NA3	Workflow diagram

Wherever AI is used, the trainee remains fully responsible for the output and must be able to understand and explain every part of the submission (see constraint CN9).

End of Document

This Business Information Document is issued for execution. Questions regarding interpretation should be raised during the Phase 1 clarification window.